VLSI Design Laboratory – Assignment 2A

RTL Design and Power-Aware Implementation of a Precision-Scalable 16×16 Systolic Tensor Unit

Department of Electrical Engineering

Duration	6 weeks
Team size	2 students per group
Design style	Synthesizable Verilog RTL
Target technology	65-nm CMOS standard-cell flow
Clock target	300 MHz

Context. In Assignment 1, you designed a 4×4 systolic array from the transistor level upward. You encountered the physical realities behind apparently simple digital operations: transistor count, routing, parasitic capacitance, layout hierarchy, DRC/LVS, timing degradation and post-layout verification. Assignment 2 approaches the same computational structure from the front-end direction: scalable RTL architecture, precision configurability, synthesis, timing closure and switching-power analysis.

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1 Introduction and Motivation

You will design a 16×16 systolic Tensor Unit (TU) containing 256 Processing Elements (PEs) using synthesizable Verilog RTL. Merely scaling the previous array from 16 PEs to 256 PEs is not the objective. The new PE shall support dynamically selectable INT4, INT8 and INT16 arithmetic. When reduced precision is selected, circuitry unnecessary for that precision should, as far as possible, cease switching.

The resulting accelerator therefore trades numerical precision against switching activity, energy consumption and potentially application accuracy. The central engineering question is:

> **If an application does not require 16-bit arithmetic, why should a 16-bit datapath continue consuming dynamic power as though it did?**

Students will investigate this question through architecture, RTL design, verification, synthesis, timing analysis and switching-activity-based power estimation.

2 Learning Objectives

At the completion of this assignment, students should be able to:

- design a moderately large computational IP at RTL;
- translate a previously understood PE architecture into synthesizable Verilog;
- construct a 16×16 systolic array using hierarchical RTL;
- implement signed INT4, INT8 and INT16 arithmetic;
- design dynamically configurable datapaths;
- understand operand isolation, clock gating and data gating;
- distinguish output masking from genuine switching-activity reduction;
- develop self-checking verification environments;
- use parameterisation and `generate` constructs appropriately;
- synthesise RTL to a standard-cell library;
- identify critical paths and interpret synthesis timing reports;
- estimate switching activity and dynamic power;
- evaluate precision-performance-power trade-offs quantitatively; and
- document a digital IP block in an engineering design report.

3 Overall Design Requirement

Design and verify a

$$\boxed{16 \times 16 \text{ Precision-Scalable Systolic Tensor Unit}}$$

consisting of 256 Processing Elements arranged as $PE_{0,0}$ to $PE_{15,15}$. Each PE shall perform signed multiply-accumulate operation of the form

$$ACC_{ij} \leftarrow ACC_{ij} + A_{ik}B_{kj}.$$

Operands shall propagate through the array systolically:

- A operands propagate from left to right;
- B operands propagate from top to bottom;
- each PE retains its local accumulated result.

The array shall support signed integer matrix multiplication

$$C = A \times B, \qquad C_{ij} = \sum_{k=0}^{15} A_{ik} B_{kj}.$$

4 Precision Modes

Two global mode signals shall control the arithmetic precision of the entire array.

`mode_1`	`mode_0`	Operating Mode
0	0	INT4
0	1	INT8
1	0	INT16
1	1	Reserved

The mode shall apply simultaneously to all 256 PEs. The selected precision shall remain constant during one matrix multiplication. The reserved state shall not accidentally initiate arithmetic. Teams may either treat it as an illegal state or propose a useful additional operating mode for bonus credit.

5 Processing Element Architecture

Every PE shall contain, conceptually:

1. an A-operand register;
2. a B-operand register;
3. signed multiplication hardware;
4. an accumulator;
5. operand forwarding registers or logic;
6. precision-control logic; and
7. switching-suppression logic.

The physical operand interface shall support 16-bit signed operands. The PE shall interpret these according to the selected mode:

- INT4: signed 4-bit operands;
- INT8: signed 8-bit operands;
- INT16: signed 16-bit operands.

Correct two's-complement sign handling is mandatory.

6 Datapath Width and Accumulator Design

Students shall derive the required multiplication and accumulation widths for each precision mode. For a 16-element dot product,

$$C_{ij} = \sum_{k=0}^{15} A_{ik} B_{kj}.$$

Your report shall derive the minimum accumulator width required to prevent overflow for INT4, INT8 and INT16. The hardware may employ one sufficiently wide physical accumulator or a segmented accumulator architecture.

The chosen implementation shall be justified in terms of correctness, area, switching activity, timing and design complexity.

> **Simply selecting the lower bits of a continuously switching full-width accumulator is not considered a power-saving architecture.**

7 Precision-Scalable Datapath

The central design problem in this assignment is the suppression of unnecessary switching. A conceptual 16-bit datapath may be viewed as:

$$\underbrace{[15:8]}_{\text{INT16 only}} \quad \underbrace{[7:4]}_{\text{INT8/INT16}} \quad \underbrace{[3:0]}_{\text{INT4/INT8/INT16}} .$$

INT4 Mode

Only circuitry necessary for 4-bit computation should remain active.

INT8 Mode

The lower 8-bit datapath is active while circuitry used only by INT16 should remain inactive where practicable.

INT16 Mode

The complete datapath is active.

Students are deliberately not given one mandatory implementation. Possible techniques include operand isolation, clock enables, clock gating, segmented registers, segmented adders, segmented multipliers, conditional operand propagation, data gating, hierarchical enable signals and other synthesisable techniques proposed by the team. Innovation is encouraged.

Internal tri-state logic should be used only if its synthesis implications are understood and justified.

8 Important Warning: Masking Is Not Power Management

The following is not sufficient:

```
product = a * b;

if (mode == INT4)
    result = product[7:0];
```

Although the output has been restricted to eight bits, the complete multiplier may still toggle internally. Similarly, masking unused accumulator output bits does not necessarily prevent the underlying arithmetic hardware from switching.

Masking an output is not power management.

Teams must demonstrate through architecture, synthesis results and switching/power analysis that reduced-precision modes actually reduce unnecessary switching activity.

9 Systolic Array Architecture

The complete accelerator shall consist of a regular 16×16 array:

256 PEs.

Students should use hierarchical RTL construction rather than manually instantiate 256 unrelated blocks. Use of parameters, arrays, `generate` loops and reusable PE modules is strongly encouraged.

A recommended hierarchy is:

```
tensor_unit_16x16
       |
       +-- control / input scheduling
       |
       +-- systolic_array_16x16
              |
              +-- PE
                  |
                  +-- operand registers
                  +-- precision control
                  +-- multiplier
                  +-- accumulator
                  +-- forwarding logic
```

Equivalent well-justified hierarchies are acceptable.

10 Input Scheduling

Input matrices shall be skewed so that A_{ik} and B_{kj} meet at PE_{ij} during the correct cycle. Students shall derive the corresponding schedule for the 16×16 array.

For PE coordinate (i, j) and inner-product index k, the matching operands shall meet at approximately

$$t = i + j + k, \qquad 0 \le i, j, k \le 15.$$

The report must include:

- a timing diagram for at least a 4×4 subsection;
- the general scheduling equation for the 16×16 case;
- latency from `start` to final result;
- array filling time;
- useful computation interval; and
- array draining time.

Do not manually list hundreds of operand values cycle by cycle.

11 Control Interface

The minimum top-level control interface shall include:

```
clk
reset
start
mode_1
mode_0
input_valid
busy
done
```

Suitable operand and result interfaces shall be designed by each team. The interface shall be clearly documented. `mode_1` and `mode_0` shall be sampled at the beginning of a computation and shall remain unchanged until `done`. A new operation need not be accepted while `busy` is asserted.

12 RTL Design Requirements

All submitted RTL shall be synthesizable. Students shall:

- use synchronous design practices;
- avoid arbitrary simulation delays;
- avoid unintended latches;
- avoid combinational loops;
- use non-blocking assignments for sequential logic;
- implement reset behaviour consistently;
- maintain clear module hierarchy;
- avoid unnecessary duplication of RTL;
- document parameters and interfaces.

13 Verification

Verification constitutes a major part of this assignment. A self-checking testbench shall be developed. The RTL result shall be automatically compared against a behavioural reference computation.

The following shall be tested in all three precision modes:

- zero matrices;
- identity matrices;
- positive operands;
- negative operands;
- mixed-sign operands;
- maximum positive values;
- maximum negative values;
- random matrices.

The testbench shall automatically report mismatches. Manual inspection of hundreds of output values in waveforms is not an acceptable verification methodology.

14 Mode Verification

Teams shall specifically demonstrate correct switching between

$$\text{INT4} \rightarrow \text{INT8} \rightarrow \text{INT16}.$$

The same numerical experiment should, wherever possible, be executed in different precision modes. Students shall explain sign extension, overflow behaviour, accumulator width, truncation if any, and mode-dependent datapath activation.

15 Synthesis

The complete 16×16 array shall be synthesised using the laboratory's standard-cell synthesis flow and available 65-nm technology library.

Report at minimum:

- total standard-cell area;
- sequential cell count;
- combinational cell count;
- inferred arithmetic structures;
- critical path;
- worst-case delay;
- maximum achievable clock frequency.

Students shall inspect synthesis warnings. A design that produces a netlist is not necessarily a correctly synthesised design.

16 Timing Requirement

The complete design shall target a minimum operating frequency of

$$\boxed{f_{CLK} = 300 \text{ MHz}}$$

corresponding to a nominal clock period of approximately

$$\boxed{T_{CLK} = 3.33 \text{ ns}}.$$

Timing closure shall be demonstrated after synthesis using the available 65-nm standard-cell library.

Students shall:

- identify the critical path;
- determine whether the dominant delay arises from multiplication, accumulation, operand forwarding, control logic, fanout or register-to-register paths;
- report setup slack at 300 MHz;
- report the maximum achievable clock frequency;
- inspect synthesis timing reports and identify the cells and logic levels contributing most strongly to the critical path;
- apply suitable optimisation techniques where required.

Possible optimisation approaches include pipelining, register retiming, restructuring arithmetic logic, reducing combinational depth, improving enable/control fanout, hierarchical buffering, alternative multiplier structures and segmentation of arithmetic datapaths.

Additional pipeline stages are permitted provided that functional behaviour remains correct, systolic scheduling is modified consistently, latency is clearly documented and `done` remains correctly aligned with output validity.

A design that fails to meet 300 MHz may still receive substantial credit if it demonstrates a technically sound architecture, complete timing analysis and a convincing explanation of the limiting path. Higher frequencies may be reported as an additional result, but shall not compensate for incorrect functionality or incomplete verification.

17 Switching Activity and Power Analysis

This is a mandatory part of the assignment. For comparable workloads, measure or estimate switching activity and dynamic power in INT4, INT8 and INT16 modes.

The same or statistically equivalent operand sequences shall be used for meaningful comparison. Where supported by the tool flow, students should use switching information obtained from RTL or gate-level simulation, such as VCD/SAIF activity.

At minimum report:

$$P_{\text{INT4}}, \qquad P_{\text{INT8}}, \qquad P_{\text{INT16}}.$$

Absolute power numbers alone are insufficient. Report normalized values such as

$$\frac{P_{\text{INT4}}}{P_{\text{INT16}}} \qquad \text{and} \qquad \frac{P_{\text{INT8}}}{P_{\text{INT16}}}.$$

18 Demonstrating Precision Gating

Each team shall provide evidence that its proposed precision-control mechanism works. Suitable evidence may include switching-activity reports, power reports, selected internal waveforms, clock-gating reports, synthesis structure and activity-factor comparisons.

Particular attention should be paid to the upper portions of operand registers, multiplier, adder, accumulator and forwarding registers.

The report shall explicitly answer:

What hardware stops switching in INT4 and INT8 modes, and why?

19 Precision-Power-Performance Study

Each team shall produce a summary table of the following form:

Parameter	INT4	INT8	INT16
Operand precision	4	8	16
Active datapath width	–	–	–
Maximum frequency	–	–	–
Dynamic power	–	–	–
Normalised power	–	–	1.00
Energy/operation or Energy/MAC	–	–	–
Numerical error/accuracy*	–	–	Reference

*Where an application-level experiment is performed.

Students shall discuss why silicon area is essentially fixed even though active hardware and switching activity change with precision.

20 Neuromorphic / AI Interpretation

The accelerator should be considered as a simplified Tensor Unit suitable for neural-network arithmetic. Different neural-network layers may tolerate different numerical precision. A future accelerator might select precision on a layer-by-layer basis to trade inference accuracy, arithmetic throughput, dynamic power and energy per inference.

Implementation of a complete neural network is not mandatory in this assignment.

21 Scope for Innovation

Teams are encouraged to explore alternative implementations of the precision-scalable PE. Examples include:

- operand-isolated multiplier;
- segmented multiplier;
- gated accumulator;
- segmented accumulator;
- independently enabled register slices;
- alternative multiplier architecture;
- local clock enables;
- hierarchical clock gating;

- alternative mode encoding;
- pipelining for improved frequency;
- zero-skipping or exploitation of operand sparsity.

A more complicated architecture does not automatically receive higher marks. Innovation shall be judged by measured benefit and engineering justification.

22 Six-Week Work Plan

Week 1 – Architecture and Reference Model

Complete PE specification, precision-width derivation, accumulator-width calculation, interface definition, 16×16 systolic scheduling, behavioural reference model and initial verification plan.
Milestone: Architecture Review.

Week 2 – Configurable PE

Complete INT4, INT8 and INT16 operation; mode selection; multiplier; accumulator; forwarding registers; and initial precision-gating mechanism.
Milestone: One PE passes all three precision modes.

Week 3 – 16×16 Array

Complete hierarchical array generation, systolic connections, input scheduling, control FSM, `start`/`busy`/`done`, and complete matrix multiplication.
Milestone: 16×16 RTL functionally operational.

Week 4 – Verification

Complete self-checking testbench, corner cases, signed arithmetic tests, random matrices, all three modes and regression testing.
Milestone: Functional Verification Review.

Week 5 – Synthesis and Timing

Complete 65-nm synthesis, area analysis, timing analysis, critical-path identification, RTL optimisation and 300 MHz timing closure.
Milestone: Synthesis and Timing Review.

Week 6 – Power and Engineering Analysis

Complete switching-activity generation, INT4/INT8/INT16 power comparison, precision-gating verification, energy comparison, final optimisation and final report.
Milestone: Final Design Review and Submission.

23 Team Organisation

Each group shall consist of two students. Both students are responsible for the complete design.

A suggested division is:

Student A

Primary responsibility for PE arithmetic, precision scaling, multiplier/accumulator architecture, synthesis and timing.

Student B

Primary responsibility for array integration, control and scheduling, verification infrastructure, and switching/power analysis.

This division does not imply independent projects. Both students must understand the complete system and shall jointly review and integrate all RTL.

24 Deliverables

Each group shall submit the following.

24.1 RTL Package

Containing PE RTL, arithmetic modules, 16×16 array, control logic, top-level module and all supporting RTL.

24.2 Verification Package

Containing self-checking testbench, test vectors, reference model, simulation scripts and regression results.

24.3 Synthesis Package

Containing:

- synthesis scripts;
- synthesis logs;
- area reports;
- timing reports;
- critical-path report;
- setup slack at 300 MHz;
- maximum achievable clock frequency;
- identification of the dominant timing path;
- before/after optimisation timing comparison;
- documentation of any pipelining, retiming, buffering or arithmetic restructuring used to achieve timing closure;
- synthesis warnings summary.

24.4 Power-Analysis Package

Containing switching-activity data, power reports and INT4/INT8/INT16 comparisons.

24.5 Engineering Design Report

Recommended length: 15–20 pages, excluding appendices. The report shall contain architecture, PE microarchitecture, precision-gating strategy, accumulator-width derivation, systolic scheduling, verification methodology, synthesis results, timing results, power results, precision-power-performance comparison, engineering trade-offs and conclusions.

The report shall additionally contain:

- timing target of 300 MHz;
- corresponding nominal clock period of approximately 3.33 ns;
- critical-path analysis;
- setup slack at 300 MHz;
- maximum achieved clock frequency;
- timing bottleneck identification;
- optimisation steps undertaken;
- comparison of timing before and after optimisation;
- impact of pipelining or retiming on systolic scheduling and overall latency;
- final statement on whether 300 MHz timing closure was achieved.

Do not fill the report with unannotated waveform or tool screenshots.

24.6 Mandatory Timing Deliverable

Each team shall include a one-page summary table:

Item	Result
Target frequency	300 MHz
Target clock period	3.33 ns
Critical-path delay	–
Setup slack at 300 MHz	–
Maximum achieved frequency	–
Number of pipeline stages	–
Total computation latency	– cycles
Timing closure achieved	Yes / No

25 Minimum Completion Requirement

A satisfactory submission shall demonstrate:

- functional 16×16 systolic operation;
- correct signed arithmetic;
- INT4, INT8 and INT16 modes;
- self-checking verification;
- synthesizable RTL;
- successful 65-nm synthesis;

- timing analysis;
- quantitative switching/power comparison.

A design implementing three numerical modes but providing no evidence of reduced switching activity in reduced-precision operation shall be considered functionally configurable but not successfully power-aware.

26 Bonus Extensions

Bonus credit may be awarded for technically convincing extensions such as:

- useful implementation of `mode_1 mode_0 = 11`;
- per-row or per-column precision control;
- pipelined high-frequency implementation;
- automatic clock-gating inference;
- application-level DNN experiment;
- layer-dependent precision scheduling;
- improved energy/MAC;
- saturation arithmetic;
- zero-skipping;
- exploitation of operand sparsity.

Any bonus feature must first preserve correctness of the mandatory design.

27 Final Engineering Question

At the end of the assignment, your group should be able to answer quantitatively:

> **What do we gain by operating this Tensor Unit at INT4 or INT8 instead of INT16, what do we lose, and what circuitry did we design to obtain that gain?**

The quality of the assignment will be judged not merely by whether the Verilog works, but by how convincingly your architecture, verification and synthesis results answer that question.

A Engineering Design Report Template

The final report shall follow the structure below. The recommended length is 15–20 pages, excluding appendices, RTL listings and automatically generated tool reports. The report should document the engineering process, design decisions, measured results and conclusions. It should not be a collection of screenshots.

A.1 Cover Page

Include course and assignment title, team number, student names, entry/roll numbers and date of submission.

A.2 Executive Summary

Recommended length: 200–300 words. Summarise what was designed; INT4/INT8/INT16 capability; precision-gating technique adopted; 65-nm synthesis result; maximum achieved frequency; whether 300 MHz timing closure was achieved; principal power-saving result; and most important engineering conclusion. This section should contain quantitative results wherever available.

A.3 Design Specifications

Summarise the implemented specifications in one table.

Parameter	Specification	Achieved
Array dimensions	16×16	
Number of PEs	256	
Precision modes	INT4/INT8/INT16	
Technology	65 nm	
Target frequency	300 MHz	
Target clock period	3.33 ns	
Maximum frequency	–	
INT4 dynamic power	–	
INT8 dynamic power	–	
INT16 dynamic power	–	

Briefly identify any deviation from the assignment specification.

A.4 System Architecture

Present the overall architecture of the Tensor Unit. Include the 16×16 systolic-array block diagram, top-level interface, dataflow, control architecture, input-skewing mechanism and output collection mechanism. Explain how operands move through the array. Derive the relationship between PE coordinates, operand index and clock cycle.

A.5 Processing Element Microarchitecture

Provide a detailed PE block diagram. Describe operand registers, multiplier, accumulator, forwarding paths, mode-control logic and valid/control signals. Explain the operation of the PE in INT4, INT8 and INT16 modes. Do not merely reproduce RTL.

A.6 Datapath and Accumulator Width Derivation

Derive the required arithmetic widths for all three modes. Show operand ranges, multiplication-result range, worst-case 16-term accumulation and required accumulator width.

Mode	Operand Width	Product Width	Required Accumulator Width	Implemented Width
INT4	4			
INT8	8			
INT16	16			

Explain any additional guard bits, saturation or truncation.

A.7 Precision-Gating Architecture

This is a major section of the report. Explain exactly how unnecessary switching is suppressed. Identify which portions of operand registers, forwarding registers, multiplier, adder, accumulator and control circuitry remain inactive in reduced-precision modes.

Include a diagram showing active and inactive datapath regions for INT4, INT8 and INT16. State clearly:

> **What circuitry stops switching, how is it prevented from switching, and what evidence demonstrates this?**

Discuss alternative architectures considered and why the submitted architecture was selected.

A.8 RTL Organisation

Show the RTL hierarchy. Identify parameterised modules, `generate` constructs, sequential blocks and combinational blocks. Do not include complete RTL listings in the main report. Important RTL excerpts may be shown where necessary to explain a design decision.

A.9 Verification Methodology

Describe the self-checking verification environment: reference model, stimulus generation, automatic result comparison, error reporting, corner cases and random testing.

Test	INT4	INT8	INT16
Zero matrix	Pass/Fail		
Identity matrix			
Positive operands			
Negative operands			
Mixed signs			
Maximum values			
Random test 1			
Random test 2			
Random test 3			

Include representative waveforms only where they demonstrate an important behaviour.

A.10 Synthesis Results

Report synthesis using the specified 65-nm library. Include standard-cell area, combinational cell count, sequential cell count, multiplier implementation, register count and major cell categories.

Metric	Result
Total cell area	
Combinational cells	
Sequential cells	
Total registers	
Total PEs	256
Target frequency	300 MHz

Discuss the major contributors to area.

A.11 Timing Closure at 300 MHz

Timing closure shall receive a separate section.

Item	Result
Target frequency	300 MHz
Target clock period	3.33 ns
Critical-path delay	
Setup slack at 300 MHz	
Maximum achieved frequency	
Pipeline stages	
Matrix-computation latency	cycles
Timing closure achieved	Yes / No

Identify the complete critical path. State whether it occurs principally through multiplier, accumulator, multiplier-accumulator combination, forwarding logic, control logic, high-fanout enable/mode network or another path. Include the relevant timing report extract.

A.12 Timing Optimisation

Document the evolution of the design.

Design Version	Critical Delay	Maximum Frequency	Principal Change
Initial RTL			
Optimisation 1			
Optimisation 2			
Final			

Discuss techniques attempted, such as pipelining, register retiming, multiplier restructuring, arithmetic segmentation, buffering, fanout reduction and control restructuring. Unsuccessful optimisation attempts may also be reported if they produced a useful engineering conclusion.

A.13 Switching Activity and Power Analysis

Explain how switching activity was obtained. State simulation workload, number of cycles, operand statistics, activity-file format and power-analysis methodology. Use equivalent workloads for the three precision modes.

Metric	INT4	INT8	INT16
Dynamic power			
Normalised dynamic power			1.00
Energy/MAC			
Switching activity			
Active datapath width			16

Where possible, separate clock power, register power, combinational power and multiplier/accumulator contribution.

A.14 Precision-Power-Performance Analysis

Combine the principal results.

Metric	INT4	INT8	INT16
Numerical precision	4 bit	8 bit	16 bit
Dynamic power			
Relative power			1.00
Maximum frequency			
Energy/MAC			
Numerical error/accuracy*			Reference

*If an application-level experiment was performed.

Explain the engineering trade-off. Do not merely state that INT4 consumes less power. Explain why, where the saving originates and whether the measured reduction agrees with expectations.

A.15 Discussion

This section should demonstrate engineering judgement. Discuss what worked well; what limited frequency; what dominated power; whether precision gating was effective; which structures continued switching unexpectedly; the cost of supporting three precision modes; timing or area overhead of configurability; and what would be redesigned if another iteration were available.

Where appropriate, relate observations to the previous full-custom 4×4 systolic-array assignment.

A.16 Conclusions

Recommended length: 300–500 words. Summarise quantitatively functionality achieved, area, maximum frequency, timing closure at 300 MHz, INT4/INT8/INT16 power, energy savings and principal architectural trade-off.

Conclude by answering:

> **What is gained by operating the Tensor Unit at INT4 or INT8 instead of INT16, what is sacrificed, and what hardware mechanism produces the measured gain?**

A.17 Individual Contributions

Provide a concise table.

Task	Student 1	Student 2
Architecture		
PE RTL		
Precision gating		
Array integration		
Verification		
Synthesis		
Timing closure		
Power analysis		
Report		

Use P for primary responsibility and S for supporting responsibility. Both students remain responsible for understanding the complete design.

A.18 References

Cite textbooks, papers, standard-cell/library documentation, EDA documentation and other technical sources used. Use IEEE reference style consistently.

A.19 Appendices to Student Report

The following may be placed in appendices rather than the main report: selected RTL listings, complete synthesis reports, detailed timing reports, additional waveforms, regression logs, power reports and supplementary calculations. Do not attach hundreds of pages of automatically generated tool output.

A.20 Report Presentation Requirements

- Main report: 15–20 A4 pages, excluding appendices.
- Use numbered figures and tables.
- Every figure and table shall be referenced and discussed in the text.
- Axes shall include quantities and units.
- Waveforms shall be cropped and annotated to show the behaviour being discussed.
- Tool screenshots should be used only when they convey engineering information.
- Numerical results should preferably be presented in tables or plots rather than buried in prose.
- RTL shall not be used to inflate report length.
- Report conclusions shall be supported by measured or synthesised results.

The report should read as the engineering record of the development of a digital IP block, not as a diary of EDA-tool operations.